

Drivers and environmental impacts of Arctic shipping

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Abstract

Sea ice retreat is opening up Arctic shipping routes, reducing transit times and enhancing access to regions with valuable resources. However, increased shipping has a negative impact on the Arctic environment. In this Review, we discuss the drivers and impacts of Arctic shipping and outline opportunities for its sustainable development. The Arctic sea ice area in September decreased by 35.8% during 1980–2024; consequently, the number of ships entering the Polar Code Arctic area increased by 37.3% between 2013 and 2023, increasing CO₂ emissions from shipping by 6.3%. Economic incentives, governance frameworks, technology and infrastructure are also key drivers. The impacts of Arctic shipping span multiple environmental spheres, including changes in wildlife behaviour, invasive species, pollution and climate forcing. For example, ship-based emissions contributed to 560–1,100 premature deaths in the Nordic Arctic in 2015; biofouling increases the risk of invasive species by a factor of 3–20; one oil spill caused up to 300,000 bird deaths; and the concentration of microplastics near Brønnøysund is one to four orders of magnitude higher than the global average probably owing to wastewater from ships. Sustainable Arctic shipping will require stricter regulations on fuel standards and greywater release alongside improvements in navigation and emission filtration technologies.

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Introduction

Over 80% of global trade is carried by sea and global maritime export volume increased by 89% during 2000–2023¹ (Fig. 1a). The largest share of maritime exports come from Asia (35.8% in 2023), followed by Europe (23.6%) and the Americas (22.0%). Trade between Asia and Europe dominates global flows, with exports from Asia to Europe accounting for 23.3% of global maritime trade (Fig. 1b). Polar regions are warming substantially faster than the global average, owing to polar amplification mechanisms. Consequently, the extent and thickness of Arctic sea ice have rapidly declined^{2,3}, and new Arctic shipping routes have emerged^{4–7}.

Arctic shipping routes serve as a strategic passage connecting the Pacific, Atlantic and Arctic Oceans and thus represent economic opportunities for shipping between Asia, Europe and the Americas^{8–12}. For example, the Northeast passage (NEP) reduces the sailing distance from Shanghai to Rotterdam by 16.7% compared with the Suez Canal route and by 36.2% compared with the route via the Cape of Good Hope¹³ (Fig. 1c). The Northwest passage (NWP) offers a 22.2% reduction in sailing distance from Rotterdam to San Francisco relative to the Panama route¹⁴ (Fig. 1d). The retreat of sea ice and the development of Arctic shipping lanes have also drawn attention to the possibilities of resource extraction in the region (such as, fisheries) and tourism¹⁵. The development of Arctic shipping is driven by an increasingly complex interplay of diverse factors, encompassing economic incentives, governance frameworks, climate change and technological developments^{16–18}.

Protecting the Arctic environment from the potential negative environmental impacts of increased shipping is vital for preserving the region's fragile ecological balance and sustaining the Arctic's human–environment system^{19–21}. Shipping can have an impact on the environment through emissions to air and water, solid waste discharge, increased risk of biological invasions and disturbances to animal behaviour. Formulating effective and feasible measures to support the sustainable growth of Arctic shipping is challenging and must follow a systems approach to account for the various drivers, which span multiple sectors. As the importance of the Arctic region in global maritime trade continues to increase it will therefore be important to understand the driving forces and environmental impacts of Arctic shipping to inform the development of such measures¹⁸.

In this Review, we discuss the drivers and environmental impacts of Arctic shipping and pathways for a sustainable future. First, we examine trends in Arctic sea ice retreat and Arctic shipping. Then, we outline the driving forces of Arctic shipping development, including economic incentives, governance, climate change and technology and infrastructure. Subsequently, we summarize the environmental impacts of Arctic shipping, which span the atmosphere, cryosphere, hydrosphere and biosphere. Finally, we provide an outlook on the future of Arctic shipping and propose policy and technological options for its sustainable development.

Trends in Arctic shipping

Anthropogenic warming is driving a continued decrease in sea ice area in the Arctic. According to satellite observations²² the annual average surface air temperature increased by 7.8% over the Arctic during 1980–2024, and the September Arctic sea ice area decreased by 35.8% over the same period (Fig. 2a,b). Climate change is expected to intensify throughout the twenty-first century². Arctic temperatures increase substantially in projections under all future scenarios through to 2050, with trends ranging from +0.03 to +0.08 °C yr^{−1} (Fig. 2a). Correspondingly,

the annual average sea ice area is projected to continue declining, with trends of −0.05 to −0.08 million km² yr^{−1} (Fig. 2b).

The three major Arctic shipping routes are the NEP, NWP and the Transpolar Sea Route²³ (Fig. 2c). The Northern Sea Route (NSR), which is part of the NEP, has the most favourable navigability and can be used seasonally, with limited year-round navigation supported by icebreakers²⁴. The NWP, which is primarily controlled and managed by Canada, remains difficult to navigate owing to its numerous islands and complex ice conditions²⁵. Moreover, a theoretical transpolar route crossing the North Pole also exists. However, this route is expected to be the last to become viable as the central Arctic Ocean is covered by thick, multi-year ice.

As Arctic shipping routes become more accessible the number of ships in the Arctic has increased, leading to an increase in carbon dioxide (CO₂) emissions. Between 2013 and 2023, the number of unique ships entering the Arctic in September increased by 45 ships per year (Fig. 2d). Fishing vessels and general cargo ships collectively accounted for approximately 50% of the traffic. The spatial extent of shipping CO₂ emissions in the Arctic (north of 60° N) expanded substantially between the years 2013 and 2023²⁶, with ship CO₂ emissions increasing by 6.3% during the same period²⁶ driven by this increase in traffic and shrinking extent of sea ice (Fig. 2e,f).

Drivers of shipping trends

Economic incentives are the primary driver of Arctic shipping, followed by governance (Fig. 3). Moreover, climate change and technology and infrastructure provide constant driving forces, shaping the safety, economic viability and environmental sustainability of Arctic shipping. Here, we outline these different drivers, where 'drivers' specifically refers to the driving forces of Arctic shipping, whereas 'impact' is used more generally to describe the influence of a factor on something else, which is not necessarily Arctic shipping.

Economic incentives

The increasing economic integration of the Arctic and the rest of the world, trade expansion, resource extraction, fisheries and tourism have driven the growth of Arctic shipping activities. In August 2022, the NSR Development Plan (to 2035) was approved by the head of the Russian cabinet, with total financing of around US\$18 billion to support the expansion of port infrastructure, navigation and safety systems, and related logistics facilities to enhance the operational capacity of the NSR²⁷. The NSR is shorter and less expensive than alternative routes that do not pass through the Arctic, making it appealing for commercial shipping²⁸. For example, choosing the NSR to travel from Yokohama to Rotterdam is projected to reduce transit time and fuel consumption by 37% and 29%, respectively in the year 2030²⁹. These savings are projected to increase further to 43% and 37% in 2050, as sailing speeds increase and fuel consumption associated with icebreaking decreases with continued sea ice retreat²⁹. In addition, NSR shipping is estimated to produce 27% less CO₂ emissions per loop than the Suez Canal Route^{29,30}. Consequently, carbon taxation policies aimed to facilitate a transition to a low-carbon economy further enhance the cost competitiveness of Arctic routes relative to traditional routes^{31,32}.

The extraction of economically valuable resources in the Arctic is a strong economic driver of the increase in Arctic shipping. The Arctic region is estimated to hold 13% and 30% of the world's undiscovered conventional reserves of oil and natural gas, respectively^{33,34}. Some of the largest Arctic natural gas reserves are found in the South Kara Sea, which is adjacent to the Russian gas production hub on the Yamal

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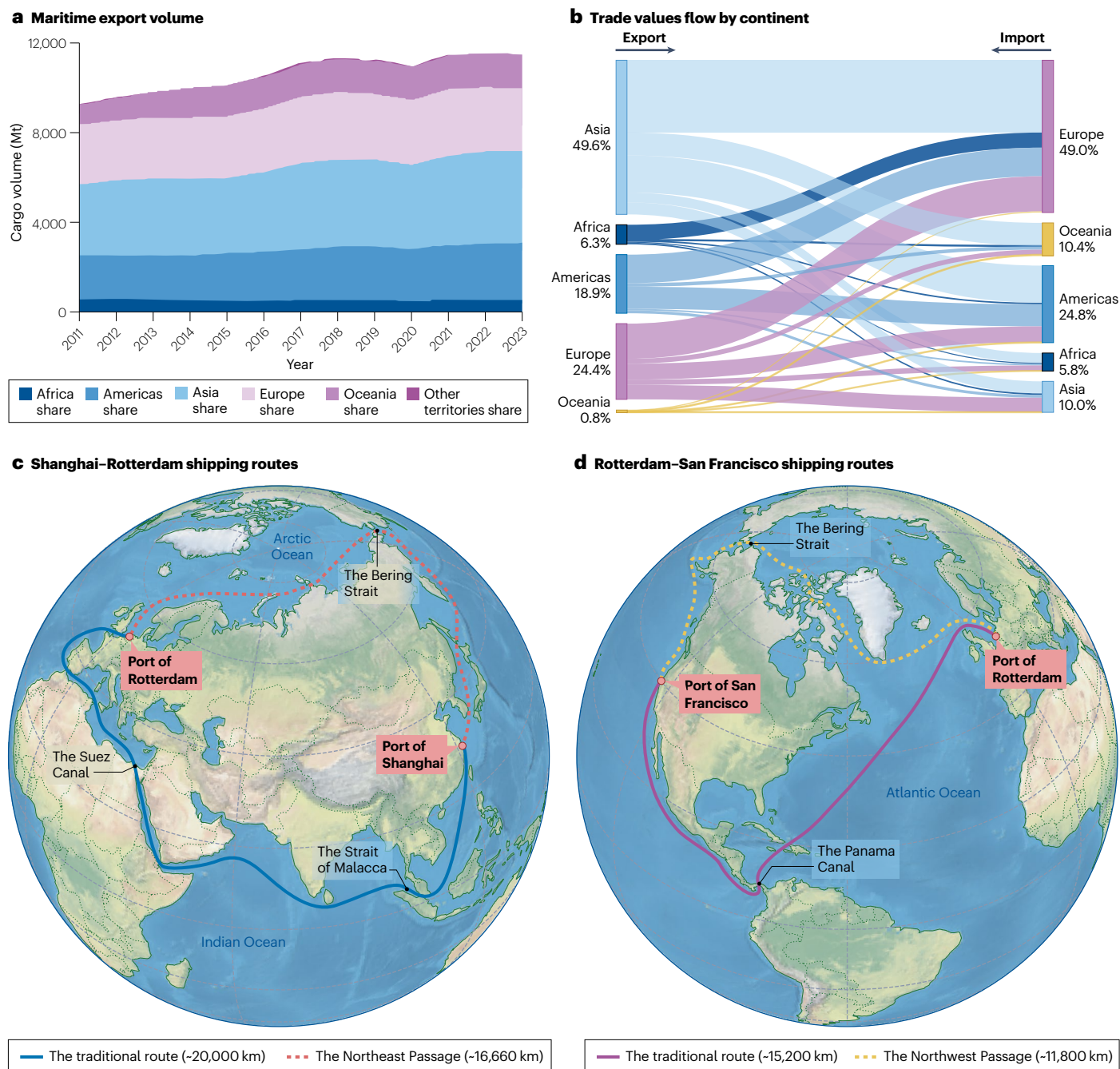


Fig. 1 | Global maritime trade and the Arctic routes. **a**, World seaborne trade export volume by economies. Cargo volumes for ‘Other territories’ are too small to be seen (0.00–0.14%, 2000–2023). Data from ref. 226. **b**, Global annual seaborne trade value shares by continent in 2024, depicting export (left) and import shares (right). **c**, A comparison of the Shanghai–Rotterdam route via the

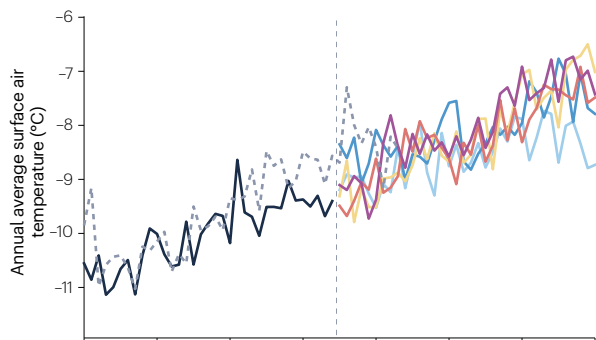
Suez Canal versus the Northeast passage¹³. **d**, A comparison of the Rotterdam–San Francisco route via the Panama Canal versus the Northwest passage¹⁴. Arctic shipping routes are emerging as critical components in the global maritime network. Data in **b** from the UN Comtrade database, <https://comtradeplus.un.org/>.

Peninsula³⁵. The Arctic also contains substantial strategic mineral deposits, including rare earth elements, copper, phosphorus, niobium, platinum-group elements and other critical metals³³. Approximately 63% of the Arctic’s total resource base is in Eurasia and 36% in North America. The Red Dog Mine in northern Alaska accounts for

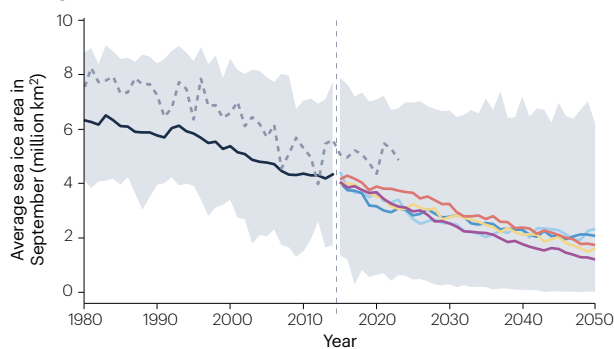
approximately 5% of global zinc production and 3% of global lead production, and between 1989 and 2011, it contributed about US\$560 million to the state’s economy³⁶. The expansion of resource extraction in the Arctic, such as the operation of the Mary River Project³⁷ since 2014 and the Yamal LNG Project³⁸ since 2017, has already led to increased

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a Surface air temperature

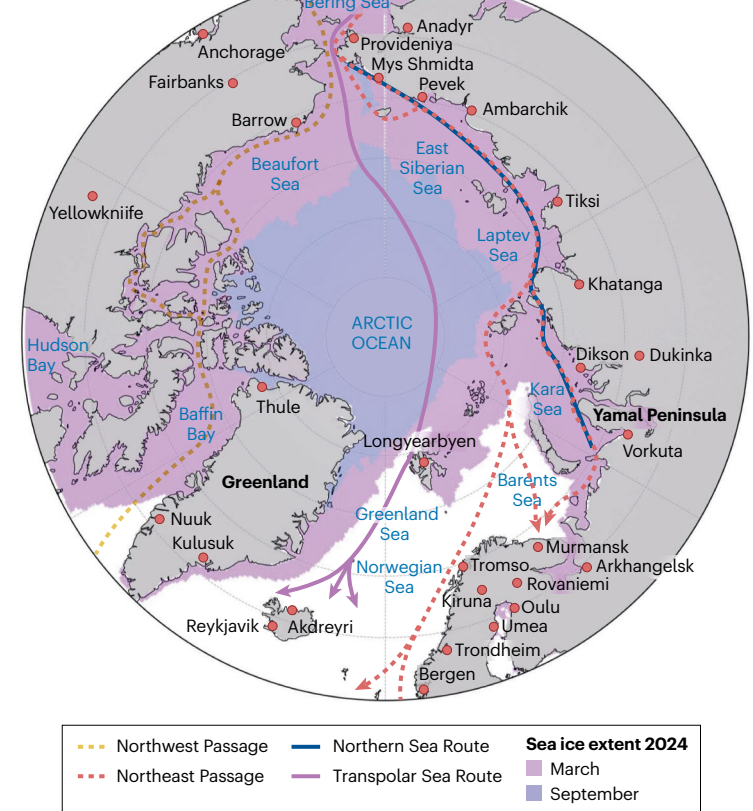


b September sea ice area



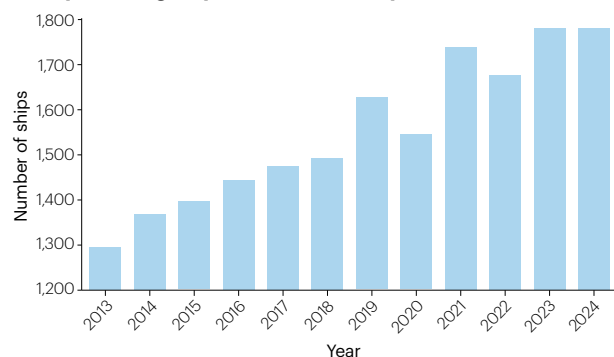
— Historical simulation — — Historical observations — Multi-model range
— SSP1-1.9 — SSP1-2.6 — SSP2-4.5 — SSP3-7.0 — SSP5-8.5

c Arctic routes



--- Northwest Passage — Northern Sea Route — Sea ice extent 2024
--- Northeast Passage — Transpolar Sea Route — March
— September

d Ships entering the polar code area in September



2013

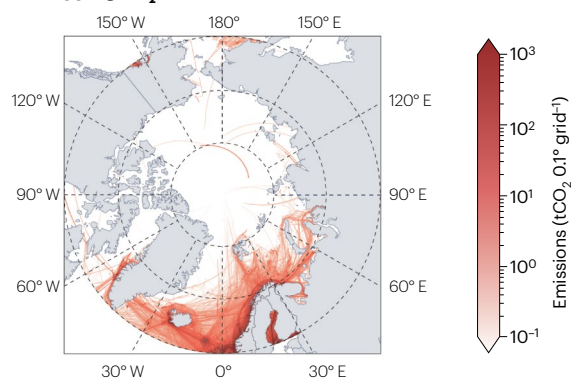


2024



■ Fishing vessels ■ Refrigerated cargo ■ Crude oil tankers
■ General cargo ships ■ Research vessels ■ Passenger ships
■ Towing or pushing ■ Oil product tankers ■ Yachts
■ Bulk carriers ■ Chemical tankers ■ Container ships
■ Other activities ■ Offshore supply ships ■ Other service offshore
■ Cruise ships ■ Gas tankers ■ Dredging vessels
■ Ro-Ro cargo ships

e Shipping CO₂ emissions in 2013



f Shipping CO₂ emissions in 2023

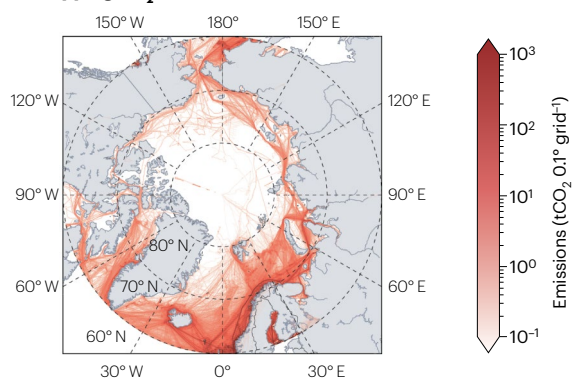


Fig. 2 | Trends in Arctic climate, sea ice and shipping. **a**, Annual average surface (2 m) air temperature over the Arctic (regions north of 60° N), based on observations (dashed grey line; 1980–2024), historical simulations (black line; 1980–2014; MPI-ESM1.2-LR²²⁷) and future projections under various shared socioeconomic pathway (SSP) scenarios (coloured lines; 2014–2050; MPI-ESM1.2-LR²²⁷). **b**, As in **a** but for average sea ice area in September over the Arctic. The shaded grey region indicates the multi-model range. **c**, Arctic routes and the sea ice extent in March and September in 2024. **d**, Left: the number of unique ships entering the Arctic in September during 2013–2024⁴¹. Right: the proportion of

each vessel type entering the Arctic in 2013 and 2024⁴¹. **e**, Shipping CO₂ emissions over regions north of 60° N in 2013²⁶. **f**, As in **e** but for 2023²⁶. The number of unique ships entering the Arctic in September increased by 45 ships per year during 2013–2023, owing to the reduction in ice caused by anthropogenic warming. Data in **a** from National Oceanic and Atmospheric Administration (<https://www.psl.noaa.gov/>); observations in **b** from refs. 228,229, Ocean and Sea Ice Satellite Application Facility (<https://www.eumetsat.int/osi-saf>); simulations in **b** from ref. 2; data in **c** from US National Ice Center (<https://usicecenter.gov/>), Arctic Portal (<https://arcticportal.org/>).

shipping traffic, particularly bulk carriers and liquefied natural gas (LNG) carriers. For example, LNG tanker activity in the Arctic was almost negligible in 2016; however, following the opening of the Yamal LNG Project, 29 LNG tankers were operating in the region by 2019, accounting for 28% of total ship fuel consumption in the International Maritime Organization (IMO) Polar Code area³⁸.

The Arctic has some of the world's most important fishing grounds, such as the Barents Sea and the Bering Sea. Commercial fisheries in Arctic and sub-Arctic waters are responsible for over 10% of the world's fish catch³⁹. In 2022, the seafood industry contributed US\$5.7 billion to Alaska's economy⁴⁰. In 2023, fishing vessels accounted for 34% of the total annual mileage of all vessel types in the Polar Code area, the highest among all vessel categories⁴¹. Climate change has also driven boreal fish communities to expand northward in regions such as the Barents Sea, potentially increasing the availability of commercially important species, such as cod and haddock, in higher-latitude waters⁴².

Arctic tourism, although accounting for only 1% of the number of vessels in the Polar Code Area (Fig. 2d), has been gradually expanding as warming temperatures and the melting cryosphere make the region more accessible. The number of tourists visiting high-latitude Arctic areas increased from 68,000 in 2008 to 98,000 in the year 2017⁴³. In addition, the number of passengers on Arctic cruises rose by over 20.5% during 2013–2018, reaching an estimated 27.2 million passengers in the year 2018⁴⁴. Popular tourist destinations include Iceland, the Svalbard Archipelago and Greenland. In 2024, cruise tourism generated NOK 360 million (US\$36.6 million) in local revenue for the Svalbard archipelago⁴⁵.

Governance

Governance constitutes the second major driver, determining divergent unilateral and multi-lateral pathways for Arctic shipping. Arctic shipping operates under a complex multi-level regulatory system, spanning international, regional and national frameworks.

At the global level, the United Nations Convention on the Law of the Sea (UNCLOS, 1982)⁴⁶ provides a fundamental legal framework for defining maritime rights, ensuring freedom of navigation and protecting the environment in Arctic waters⁴⁷. Several agreements and initiatives have built on this framework to address environmental challenges, for example, the Biodiversity Beyond National Jurisdiction (BBNJ), also known as the High Seas Treaty, 2023; <https://www.un.org/bbnjagreement/en>) Agreement establishes frameworks for marine protected areas, environmental and ecological impact assessments and sustainable ocean management. The IMO has established mandatory standards for Arctic vessel construction, operations and environmental protection through industry regulations such as Polar Operational Limitations Assessment Risk Indexing System (POLARIS, 2016)⁴⁸, Polar Code (2017; <https://www.imo.org/en/ourwork/safety/pages/polar-code.aspx>) and Arctic Ice Regime Shipping System (AIRSS, 2020; <https://tc.canada.ca/en/>

<https://www.imo.org/en/mediacentre/hottopics/pages/polar-default.aspx>)⁴⁹. The IMO also introduced a strategy to reduce greenhouse gas (GHG) emissions from ships by promoting zero or near-zero emission fuels and technologies (IMO, 2023; <https://www.imo.org/en/ourwork/environment/pages/2023-imo-strategy-on-reduction-of-ghg-emissions-from-ships.aspx>)⁵⁰. These frameworks facilitate the safety of vessels operating in the Arctic and provide a robust legal and regulatory basis for minimizing the environmental impacts of Arctic shipping.

At the regional level, the Arctic Council (<https://arctic-council.org/about/>), established in 1996, has a central role in Arctic governance⁴⁷. As a forum-based organization composed of member states and observer countries, its authority primarily lies in policy coordination and scientific collaboration rather than legally binding resolutions. The Arctic Council actively promotes environmental impact assessments related to Arctic shipping^{43,51,52}. However, the Council's regular operations have been affected by geopolitical factors. For example, 7 of the Arctic Council's 13 working groups have been disrupted by the Russia–Ukraine war^{53,54}.

At the national level, Arctic governance is shaped by various legal, regulatory, and policy frameworks⁵⁵. The eight Arctic states (Canada, Denmark, Finland, Iceland, Norway, Russia, Sweden and the USA) cooperate through multi-lateral frameworks, such as the Arctic Environmental Protection Strategy (AEPS, 1991)⁵⁶. These states have also developed their own national regulations to protect the Arctic environment and promote the sustainable development of Arctic shipping, for example, the NSR navigation rules of Russia⁵⁷, Arctic Waters Pollution Prevention Act (AWPPA) of Canada⁵⁸ and Svalbard environmental regulations of Norway^{59,60}. The participation of non-Arctic countries and regions such as China⁶¹, Japan⁶² and South Korea⁶³ in Arctic shipping and Arctic environmental governance is increasing. For example, China has introduced an Arctic Policy that aims to protect the Arctic eco-environment and mitigate climate change while also promoting the development of Arctic shipping routes⁶⁴. National and international policies formally protect the rights, lands and traditional livelihoods of Indigenous peoples^{65,66}, and require that these rights are considered and appropriate consultations performed when planning Arctic shipping activities. However, influence of Indigenous groups on Arctic shipping development remains limited⁶⁷.

Climate change

Increased melting and retreat of sea ice, fog, waves and storm conditions under climate change can act to increase or decrease Arctic shipping. The net effect of climate change on Arctic navigability depends on the season, the specific route and the vessel type.

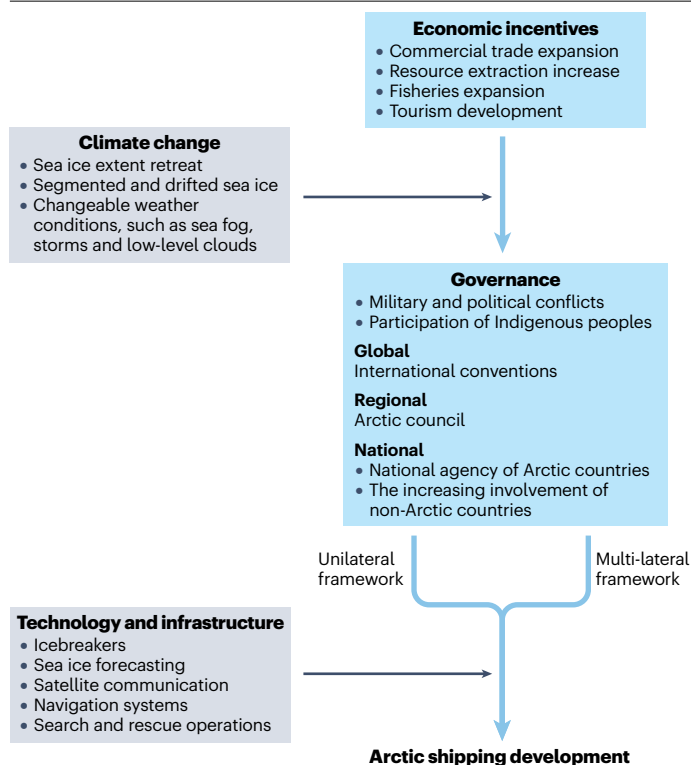


Fig. 3 | Major driving forces of Arctic shipping. A summary of the drivers of Arctic shipping development. Economic incentives are the primary driver of Arctic shipping, followed by governance frameworks, which are multi-layered and deeply intertwined with geopolitical dynamics. Climate change and technology and infrastructure provide constant driving forces, shaping the safety, economic viability and environmental sustainability of Arctic shipping. Arctic shipping development is shaped by the complex interplay of various drivers.

Retreating Arctic sea ice has extended the navigable window of shipping routes and enhanced accessibility along the NEP⁶⁸. From the 1980s to the 2010s, the navigable window of NEP for open-water (unreinforced) vessels increased from less than 10 days to 89 ± 16 days, whereas that for Polar Class 6 (roughly comparable to Arc4, the most commonly used ship type on Arctic routes) vessels expanded from no more than 60 days to 163 ± 19 days⁶⁹. Although the navigable window represents the period when ice conditions theoretically allow navigation, actual commercial transits depend on additional factors such as insurance and economic viability. Consequently, a conventional container ship did not transit through the NEP until the year 2023⁷⁰. In addition, despite basin-scale ice retreat, navigation along the NWP is constrained by persistent choke points, such as M'Clure Strait, Viscount Melville Sound and M'Clintock Channel, because of their narrow channels and lingering multi-year ice⁷¹. In addition, the melting of sea ice is also expected to increase the variability of ice drift⁷², thereby raising the risk of ship–ice collisions^{73–75}.

Increasingly changeable weather conditions owing to climate change can make it more difficult to navigate the Arctic. As sea ice coverage declines, air humidity in the Arctic increases and the ocean–atmosphere interface expands, leading to a substantial increase in the frequency and duration of sea fog and low-level clouds^{76–78}. These conditions reduce visibility and impair the performance of onboard

sensors such as radar and infrared systems, making it more difficult for vessels to detect hazards such as icebergs and drift ice⁷⁹. Climate change is also expected to lead to more frequent and intense extreme wave events⁸⁰ and the intensification of near-surface winds over the Arctic Ocean^{81–83}. The anticipated rise in extreme weather conditions, alongside persistent cold temperatures, is likely to increase physical and mental stress and safety risks for crew members^{84,85}, further complicating safe navigation in the Arctic.

Technology and infrastructure

Owing to the harsh natural conditions of Arctic routes, icebreaker support, reliable forecasts for weather and sea ice conditions, sufficient navigation infrastructure and search-and-rescue operations in remote areas are essential^{86,87}. Icebreakers, which are specially designed ships capable of navigating through sea ice by breaking it with their reinforced hulls and powerful engines, are critical for ensuring the viability of Arctic shipping, because they enable other vessels to transit ice-covered waters safely. The Russian icebreaker fleet expanded from 18 vessels in 1989 to 38 vessels by 2020, primarily powered by nuclear and diesel engines^{88–90}. Similarly, by 2022, the Canadian Coast Guard maintained a fleet of 20 icebreakers⁹¹. In 2023, Russia's nuclear icebreaker fleet escorted over 730 vessels on the NSR⁹², which increased to 976 in the year 2024^{93–95}. Moreover, in the NWP, the Canadian Coast Guard logged 65 commercial escorts in the 2022, which increased to 39 in the year 2024^{96,97}.

Sea ice and weather forecasts continue to develop, increasingly integrating new methods and technologies. The Polar Code requires ship masters to consider hazards and information limitations when planning a voyage through polar waters⁹⁸. The operational limits of a voyage are defined by a risk index outcome (computed with POLARIS), which is based on ship class and ice regime^{99–103}. Advances in remote sensing³, numerical modelling¹⁰⁴ and machine learning^{105,106} have improved the accuracy of sea ice classification and trend prediction^{105,107–109}. However, uncertainties in sea ice processes, the lack of observational data and uncertainties in initial parameters and boundary conditions of models limit the reliability of real-time sea ice forecasting near shipping routes^{107,110,111}. Arctic weather forecasting is similarly challenged by the limited ability to accurately simulate and assimilate the complex atmosphere–sea–ice–ocean processes characteristic of the region^{80,112,113}. Furthermore, the Arctic region has a very sparse network of fixed weather stations and meteorological buoys¹¹²; consequently, forecasts are less accurate than those for mid- and low-latitude regions¹¹⁴.

Communication and navigation on Arctic routes present a major technological challenge. The International Convention for the Safety of Life at Sea (SOLAS)¹¹⁵ and International Code for Ships Operating in Polar Waters (Polar Code)⁹⁸ require that ships operating in Arctic waters are equipped with satellite communication systems. Arctic shipping relies on Global Navigation Satellite Systems (GNSS) for navigation^{116–118}. However, the low elevation angles of polar-orbiting satellites, frequent ionospheric scintillation in polar regions and the limited availability of ground-based augmentation systems hinders the use of GNSS in the Arctic^{119–122}.

Ports and rescue infrastructure are needed to support sustained large-scale shipping operations^{123–127}. There has been some expansion and modernization of Arctic ports and coastal infrastructure. Specialized Arctic ports featuring deep-water terminals and extensive ice management capabilities have been constructed. For example, Sabetta on the Yamal Peninsula has enabled year-round operation

for LNG exports and serves as a critical logistical hub on the NSR¹²⁸. However, substantial shortcomings still need to be overcome, for example, efforts are needed to increase search-and-rescue coverage on main Arctic shipping lanes and the ability of ports to support emergency operations in extreme Arctic conditions¹²⁷.

Environmental impacts of Arctic shipping

Arctic shipping has an impact on the environment through animal behaviour changes and collisions, oil spills, invasive species, acoustic impacts, changes in air quality that affect human health, shipboard wastewater and waste discharges, aerosol climate forcing and snow albedo forcing. Although research on the impacts of human activities on the Arctic environment (north of 60° N) is expanding^{17,129,130}, quantitative assessments of shipping sector's impacts in the Arctic are limited. This section outlines the environmental impacts of Arctic shipping across the atmosphere, cryosphere, hydrosphere and biosphere (Fig. 4, Supplementary Tables 1–4, Supplementary Note 1 and Supplementary Fig 1).

Atmosphere

The combustion of fuels in ships generates numerous pollutants, including sulfur oxides (SO_x), nitrogen oxides (NO_x), CO₂, carbon monoxide (CO), particulate matter, hydrocarbons, methane and volatile organic compounds (VOCs). These emissions have an impact on the Arctic environment in various ways. For example, SO_x, NO_x, CO and particulate matter negatively affect air quality¹³¹. In addition, these emissions can also deposit into water bodies through dry and wet deposition processes, contributing to acidification and eutrophication^{132–134}. NO_x and VOCs undergo photochemical reactions to produce ozone (O₃), which can threaten Arctic plants and animals by inducing oxidative stress and impairing respiratory health¹³⁵. Moreover, CO₂ and methane are major GHGs driving climate warming. In the Arctic, frequent surface-based temperature inversions, weak turbulent mixing and shallow boundary layer heights reduce vertical dispersion, causing these pollutants to remain concentrated near emission sources for longer periods than in mid-latitude regions¹³⁶.

Shipping emissions make a substantial local contribution to Arctic air pollution; however, research on air quality and human health is dominated by modelling studies and remains weakly constrained by observations (Fig. 4f). Atmospheric chemistry models indicate that during 2012–2015, shipping activities increased atmospheric NO_x concentrations by 10–80%, PM_{2.5} (particulate matter with an aerodynamic diameter ≤2.5 μm) by 1–15% and O₃ by around 5% along Norwegian and Canadian Arctic waterways and coastal areas^{137,138}. It is estimated that in 2015, ship emissions in the Nordic Arctic caused 560–1,100 premature human deaths¹³⁹. As transportation activities expand, these emissions increase. Between 2013 and 2023, shipping NO_x and SO₂ emissions around the NSR increased by 115% and 68%, respectively¹⁴⁰. Arctic ship exhaust emissions are projected to increase by up to ~600% by 2050 relative to 2020, although estimates vary by region^{15,141–144}. Over 50% of global annual ship fuel consumption is HFO¹⁴⁵. Without exhaust gas cleaning systems, the SO_x emission factor of ships using HFO is five to seven times higher than that of distillate fuels, and the PM emission factor is also five to seven times higher¹⁴⁶. Thus, the use of HFO is an ongoing environmental burden in the Arctic.

Cryosphere

Emissions from Arctic shipping affect sea ice, with implications for all physical and biogeochemical processes at the Arctic air–ice–ocean

interface¹⁴⁷. Black carbon emissions deposit on snow and the ice surface, reducing the albedo, thereby accelerating snowmelt and sea ice loss. The warming effect of black carbon in the Arctic is nearly five times greater per unit mass than in mid-latitude regions, contributing to approximately 20–25% of Arctic warming during the years 1980–2018^{52,148}. The climate effects of ship-emitted black carbon accelerate sea ice melt and, in turn, create potentially greater shipping opportunities, reinforcing a positive feedback loop^{29,149,150}. In 2004, approximately 0.3% of black carbon deposition on Arctic snow and ice was attributed to Arctic shipping. This proportion is projected to increase to up to 0.7% by 2050¹⁴⁹.

Other ship emissions, including SO_x, NO_x, CO, particulate matter and VOCs form aerosols through atmospheric chemical processes, which indirectly affect Arctic sea ice via aerosol–radiation and aerosol–cloud–sea ice interactions^{135,151–153}. Aerosols typically exert a cooling effect, in contrast to the warming effect of black carbon deposition. For example, sulfate aerosols can increase cloud droplet number concentrations and cloud albedo, reducing the amount of solar radiation reaching the surface, which enhances sea ice extent^{2,135}. Thus, the opposing climate effects of different ship-emitted air pollutants can cancel each other out²⁹.

Although the Arctic is more climatically sensitive than the global average, the overall climate impact of shipping-related GHG and aerosol emissions remains negligible^{29,148,149,154,155}. Climate models suggest that aerosol climate forcing and snow albedo forcing of Arctic ship emissions are on the order of 10^{−4} W m^{−2} (refs. 29,148,149,154,156). This radiative forcing effect is projected to remain marginal even under a tenfold increase in Arctic shipping emissions¹⁴⁸.

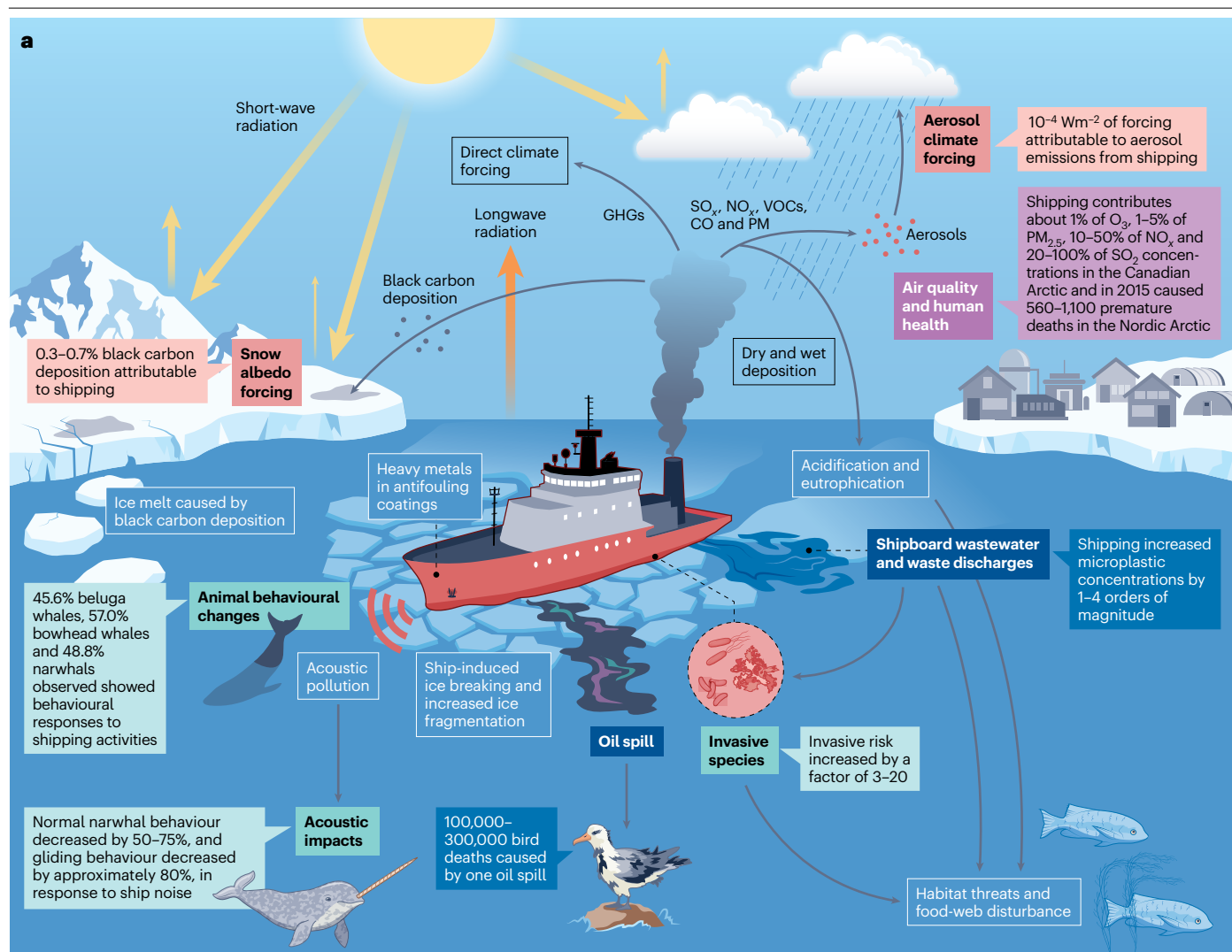
Hydrosphere

Arctic shipping has an impact on the region's hydrosphere through fuel oil leaks, wastewater discharge, marine litter and microplastic emissions. This section focuses on the impact on water quality to avoid overlap with the discussion of biospheric impacts.

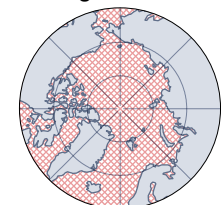
Wastewater discharge from ships can release various pollutants into water bodies and marine ecosystems. Aquatic pollutants in the Arctic often accumulate rather degrade owing to the behaviour of the ocean currents, low temperatures and weak sunlight. Consequently, the concentrations of the pharmaceuticals and personal care products^{157,158}, persistent organic pollutants¹⁵⁹ and microplastics^{160,161} are higher in the Arctic than in many other regions of the world. Greywater (used water from sinks, laundry machines, bathtubs, shower-stalls or dishwashers) discharge from ships increased microplastic concentrations by one to four orders of magnitude in Norway's coastal waters around the year 2020^{162,163}.

Quantitative understanding of the contribution of Arctic shipping to aquatic pollution remains limited (Fig. 4g). There are few measurements of the release of pharmaceuticals, personal care products and persistent organic pollutants from Arctic shipping emissions. Moreover, research rarely distinguishes between microplastics from Arctic shipping and those transported via atmospheric or oceanic circulation^{161,164,165}. Consequently, the attribution of microfibrils to shipboard wastewater and waste discharges remains highly uncertain, resulting in a comparatively low confidence level of evidence. Measurements of shipboard wastewater and waste discharges in the Arctic are limited to the Norwegian coastal ports, the northern Bering Sea and Svalbard, and coverage needs to be extended to broader areas^{163,166,167}. Quantitative assessments of acidification and eutrophication caused by the deposition of emissions from Arctic shipping are also rare, but

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b Animal behavioural changes



c Oil spill



d Invasive species



e Acoustic impacts



f Air quality and human health



g Shipboard wastewater and waste discharges



h Aerosol climate forcing



i Snow albedo forcing



Severity of impact

■ High
■ Moderate
■ Low

Confidence of evidence

■ High
■ Moderate
■ Low

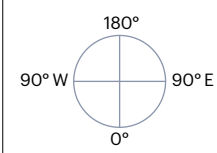


Fig. 4 | Multi-sphere impacts of Arctic shipping. **a**, A schematic summarizing the impacts of Arctic shipping on the cryosphere, biosphere, hydrosphere and atmosphere. The light-coloured boxes outline specific examples of ship-induced animal behaviour changes¹⁷⁶ and the impacts of oil spills¹⁶⁸, invasive species¹⁸⁵, acoustic noise¹⁸², air pollution^{137–139}, shipboard wastewater¹⁶³, aerosol climate forcing^{29,148,155} and snow albedo forcing^{148,149,154}. **b–i**, The geographic coverage of existing literature, the severity of the measured impact and the confidence of available evidence for animal behavioural changes (**b**), oil spills

(**c**), invasive species (**d**), acoustic impact (**e**), air quality and human health (**f**), shipboard wastewater and waste discharges (**g**), aerosol climate forcing (**h**) and snow albedo forcing (**i**) in the Arctic region north of 60° N (see Supplementary Tables 1–4 and Supplementary Note 1 for details). Arctic shipping has brought pronounced cross-sphere environmental impacts to the Arctic region. GHG, greenhouse gas; NO_x, nitrogen oxides; PM_{2.5}, particulate matter with an aerodynamic diameter ≤2.5 µm; SO_x, sulfur oxides; VOC, volatile organic compounds.

these processes can also have an impact on aquatic ecosystems by altering water pH, nutrient availability and the structure of microbial and algal communities^{132–134}.

Oil spills from ships in the Arctic can result in severe ecological disasters^{168–171}. A single major spill can cause the mortality of 300,000–500,000 seabirds^{168,170–172}. However, evidence of the impact of oil spills is concentrated in areas with a history of shipping-related accidents (Fig. 4c). Only a small portion of oil-degrading microorganisms adapt to the low temperatures of the Arctic; therefore, 1.8–2.7% of fuel oil leaked by Arctic shipping is degraded each year¹⁷², compared with ~6% in warmer waters along the Atlantic coasts of France and Spain¹⁷³. This slow degradation rate reinforces the need to ban the use of HFO in the Arctic, as it forms thicker, longer-lasting slicks than other fuel types owing to its high viscosity and persistence¹³⁷.

Biosphere

Arctic shipping has an impact on the biosphere through chemical pollution, physical disturbances, biological invasions and ecosystem alterations. In terms of chemical pollution, Arctic species exhibit distinct characteristics and adaptations to chemical pollutants^{174,175}. For example, copper, which is widely used in shipping as an antifouling coating, can affect the reproductive development of Arctic copepods and even cause death, with Arctic copepod species being more sensitive to copper ions than their temperate counterparts¹⁶⁶.

Animal behavioural change, in response to Arctic shipping, through collisions, noise and other pathways (for example, attraction to fishing gear), have been observed across the whole Arctic in multiple large marine mammal species (Fig. 4b). Over 40% of observations report that Arctic shipping (including ships, ship-related equipment and ship-related infrastructure) has a pronounced impact on animal behaviour, including changes in movement and spatial distribution, activity patterns, diving behaviour and foraging behaviour^{176,177}. Around 45% (139 of 305) of records of Arctic-specific population belugas from 2014 to 2023 have directly observed behavioural responses to ships and their associated activities¹⁷⁶. Icebreakers within about 2 km and tankers up to roughly 52 km away have the potential to substantially affect marine mammals^{178,179}. Increases in shipping traffic are expected to drive a 5-dB rise in underwater noise levels in the Arctic Ocean by 2030, corresponding to nearly a fourfold increase in sound intensity¹⁸⁰. Exposure to such vessel sounds, especially those accompanied by airgun pulses, reduces the deep-diving and foraging activities of narwhals by 50–75%^{181,182}. Much of the evidence for this reduction in normal behaviour in response to ship noise is localized near Greenland^{181–183} (Fig. 4e).

In addition, as the number of ships in the Arctic grows, the risks of ship strikes on large whale species continues to rise¹⁸⁴. These collisions not only threaten the welfare of whales but also pose risks to mariners, including vessel damage and crew injuries, particularly for smaller vessels^{177,184}. Species such as the bowhead whale, an Arctic-endemic species, are particularly at risk from collisions because the period

when they are most active (July–October) coincides with the navigable window of Arctic shipping routes¹⁷⁷.

Ships typically carry invasive species¹⁸⁵, and biofouling on ships and ballast water discharge can introduce non-native species to local environments^{186,187}. Invasive species can have an impact on the Arctic ecosystem through habitat fragmentation and loss, as well as alterations in food web^{188,189}. The seven highest risk invasive species are most likely to arrive in the Arctic Svalbard Islands as biofouling on ship hulls¹⁶². Shipping is estimated to increase the risk of invasive species by a factor of 3–20 in the Canadian Arctic and near the Svalbard archipelago, relative to a hypothetical scenario with no shipping^{162,185,188,190} (Fig. 4d). The number of pathways for the introduction of potential invasive species increases nonlinearly with the number of voyages and connectivity of global Arctic routes^{190,191}.

The growth of Arctic shipping has also had an impact on Indigenous peoples. Increased shipping traffic disrupts Arctic wildlife migration and foraging¹⁹² and declining sea ice and increased icebreaking make traditional hunting more dangerous^{193,194}. Pollutants and greywater and waste discharges from shipping also threaten the health of Indigenous people^{178,192,195}. However, there is a lack of quantitative assessments of the direct impacts of shipping on Indigenous peoples, as most research either does not specifically address shipping or consists primarily of policy reviews and interviews.

Prospects for Arctic shipping

The dynamic interplay between the drivers, development and impacts of Arctic shipping can be summarized by a ‘driver–development–impact–driver–development’ chain (Box 1). Leveraging driver-based regulation will be essential to promote the sustainable development of Arctic shipping. This section outlines how the drivers of Arctic shipping are projected to evolve and proposes measures to modify drivers to ensure the sustainability of future Arctic shipping.

Projected changes in drivers

The economic incentives for Arctic shipping are expected to grow under ongoing climate change, as Arctic accessibility increases. As Arctic shipping activities expand, conflicting sectoral interests and limited environmental carrying capacity are likely to intensify competition among stakeholders^{196,197}. For example, the tourism sector has voiced concerns that resource extraction undermines the pristine natural environment, reducing the region’s attractiveness for visitors¹⁹⁸. The uneven distribution of natural resources, with natural gas predominantly located in the Eurasian Arctic Basin and oil mainly found in the North America Arctic Basin³³, is also likely to lead to an increase in conflicts of interest between different countries. These conflicts of interest could influence the future development of Arctic shipping in several ways. For example, tensions among sectors and countries might lead to stricter environmental regulations and more complex permitting processes regarding the use of Arctic shipping routes. Moreover,

economic and geopolitical uncertainty caused by such conflicts might also make international investors more cautious.

Arctic sea ice is projected to continue retreating throughout the century, leading to increased navigability. Arctic sea ice annual mean extent and ice volume are projected to decline by about 15% and 40%, respectively, by 2100 under the 2 °C warming scenario outlined by the Paris Agreement^{199–201}. Under this scenario Polar Class 6 and open-water vessels are expected to be able to navigate the NEP for about 220 and 120 days per year, respectively, after 2040^{24,202}. In addition, in the NWP, under the 2 °C warming scenario, navigation windows are projected to reach about 200 days for Polar Class 6 vessels and 100 days for open-water vessels around 2060^{202–205}.

However, there are some uncertainties in projections of the future impacts of climate change on Arctic shipping. Under a high-emission scenario (SSP5-8.5), summer sea ice is likely to vanish almost

entirely by 2050, with a sustained ice-free summer expected by the 2030s^{199,206,207}. Moreover, under this scenario it is projected that ice-strengthened vessels will be able to operate in the NEP and NWP throughout the year by 2070²⁴. The number of navigable days after 2050 projected for the higher emissions scenario is substantially higher than projections under the Paris Agreement scenario, although the difference before 2050 remains small because substantial sea ice decline does not occur until mid-century under either scenario²⁴. In addition, climate change is also expected to increase operational hazards such as sea ice drift, extreme waves and sea fog, leading to some uncertainty in the overall increase in the accessibility of Arctic routes^{71,208}.

Geopolitics can also shape the future of Arctic shipping through multi-lateral and unilateral governance frameworks. Geopolitical instabilities are expected to shape the development of Arctic shipping. For example, the Russia–Ukraine war is likely to influence whether the NSR evolves into an internationally used global transit corridor or is increasingly confined to a Russian resource transportation corridor^{209–211}. If access to Arctic routes is governed by unilateral policies (for example, Russia or Canada impose restrictions on route usage), the outcome would probably be a fragmented system in which national actors negotiate trade-offs between sectoral interests and environmental protection within their own jurisdictions. By contrast, a multi-lateral governance framework could facilitate cooperation and enable the shared development of safer and greener Arctic shipping corridors.

In addition, environmental considerations are increasingly acting as a driver of Arctic shipping as market-based mechanisms such as carbon pricing reshape route economics. For example, the EU Emissions Trading System, which requires shipowners to hold allowances for GHG emissions along their voyages, could incentivize more trans-Arctic voyages, because Arctic routes can reduce GHG emissions compared with traditional routes²¹². However, the impact of these mechanisms on the cost-effectiveness of Arctic routes remains uncertain.

Sustainable opportunities for Arctic shipping

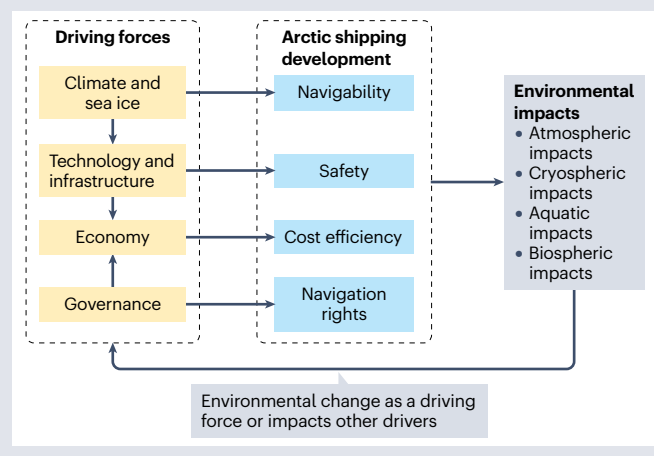
The integration of technology, economic tools, and governance is essential for sustainable Arctic shipping^{18,51,213–215} (Box 2 and Supplementary Table 5). In terms of technology and infrastructure, future developments should address two dimensions: safety and sustainability. To enhance the safety of Arctic shipping, efforts to improve real-time sea ice and weather forecasting systems, enhance satellite-based communication and navigation systems, develop port infrastructure capable of operating under Arctic extreme conditions (such as, ice-resistant facilities and cold-adapted logistics systems) and establish rapid emergency response networks are essential and urgent.

Technology is rapidly developing to ensure the sustainability of Arctic shipping and minimize its environmental impacts. Established technological measures include selective catalytic reduction systems that reduce NO_x emissions from ships, greywater treatment technologies and speed reduction measures and route optimization planning to reduce the risk of ship strikes and mitigate the impact on animal behaviour. In addition, strengthening biofouling monitoring and cleaning could help to lower the risk of the introduction of invasive species. Minimizing environmental pollution will require the development of clean energy supply and refuelling systems for Arctic ships. The technological maturity of alternative energy supply systems remains low. The development of shore power infrastructure to use the Arctic's abundant renewable energy resources such as wind, solar and hydropower, holds strong potential for decarbonizing shipping and also for providing an electricity supply for other sectors^{216–219}.

Box 1 | The driver–development–impact–driver–development chain

This Box outlines the suggested ‘driver–development–impact–driver–development’ chain (see figure), which provides a conceptual framework for Arctic shipping development.

The primary drivers of Arctic route development include economic incentives, governance, climate change and technology and infrastructure. Among these drivers, economic incentives serve as the dominant force, whereas the other drivers either facilitate or hinder development by shaping navigability, safety, cost efficiency and navigation rights. The development of Arctic shipping leads to environmental impacts, which extend across the atmosphere, cryosphere, hydrosphere and biosphere, arising from pollutant emissions, deposition, waste discharges, underwater noise, ship strikes and other stressors. These impacts, in turn, feed back into the drivers and further influence Arctic shipping development. For instance, the melting of Arctic sea ice enhances navigability, thereby stimulating shipping activities and associated black carbon emissions, which further accelerate ice loss, linking to the climate change driver. Similarly, chemical pollution and biodiversity impacts have mobilized responses from policymakers and environmental advocates, prompting regulatory and technological interventions, corresponding to the governance and technology drivers.



Box 2 | Recommendations for sustainable Arctic shipping

As temperatures increase, shipping channels in the Arctic will become wider and increasingly free of ice fragments (see figure). Consequently, the number of ships that can pass through these routes will increase and the proportion of conventional cargo vessels in the fleet will continue to rise as sea ice declines. Recommended technologies (and their maturity) (Supplementary Table 5), economic tools and governance frameworks to support the sustainability of future Arctic shipping development are outlined in this Box.

Governance frameworks

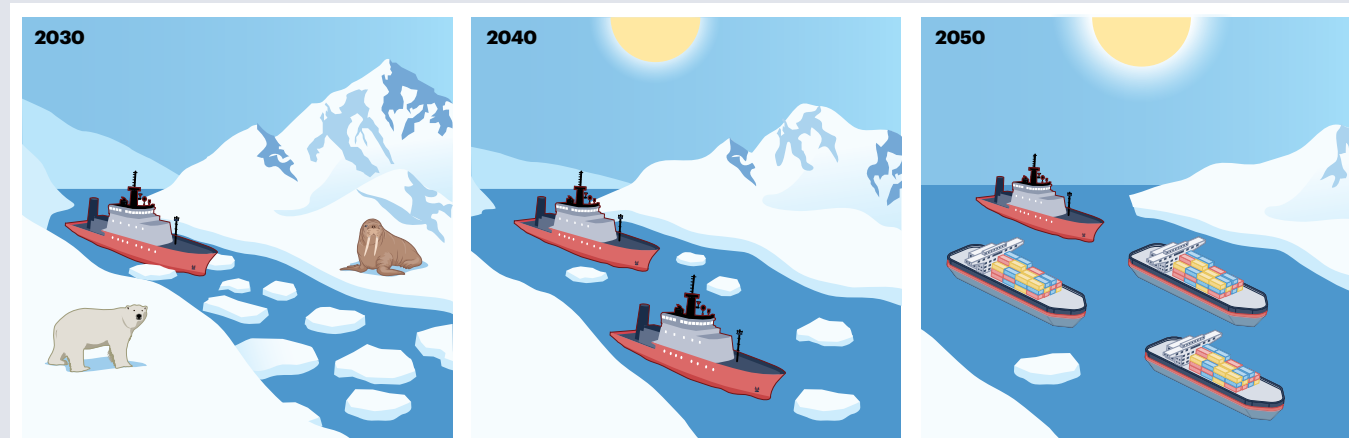
- Upgrade fuel standards, regulate the discharge of greywater and designate potential marine protected areas
- Ensure effective implementation of policies that integrate Arctic environmental protection
- International cooperation to support safe and sustainable large-scale shipping

Market mechanisms

- Facilitate a global maritime carbon trading system
- Establish environmental surcharges for Arctic navigation

Technology and infrastructure

- Increase the number and capacity of icebreakers (maturity: high)
- Improve accuracy of sea ice and weather forecasting (maturity: moderate)
- Enhance satellite communication and navigation systems (maturity: moderate)
- Develop emergency response systems and Arctic port infrastructure (maturity: low)
- Equip ships with exhaust gas after-treatment systems and wastewater treatment facilities (maturity: high)
- Apply speed restrictions (slow steaming) and route optimization (maturity: high)
- Enhance alternative energy supplies, such as shore power (maturity: moderate)
- Mitigate the invasive species risk by strengthening biofouling monitoring and implementing regular cleaning practices (maturity: high)



From a market-based perspective, economic incentives and disincentives have a crucial role in aligning the development of Arctic shipping with environmental goals. Environmental governance often follows the polluter pays principle, in which those responsible for pollution are required to bear the costs of preventing or remedying the resulting environmental damage. However, targeted market-based mechanisms for Arctic shipping remain largely absent and are rarely implemented in practice. In the short term, pilot programmes can be launched to test the use of Arctic-specific environmental surcharges or carbon pricing schemes.

In terms of governance, international cooperation is essential for ensuring the safety and sustainability of future Arctic shipping. Although existing governance frameworks support the long-term sustainability of Arctic shipping, practical actions remain insufficient. International cooperation remains indispensable for reliable large-scale Arctic shipping, as effective search-and-rescue operations, oil-spill preparedness and response, harmonized safety standards and

shared information services exceed the capacity of any single state^{129,220}. International governance is also needed to promote cleaner marine fuels, regulate greywater discharge and designate potential ecological protection zones.

Summary and future perspectives

The number of unique ships entering the Arctic in September increased by 45 ships per year during 2013–2023. This growth is shaped by the complex interplay of economic incentives, governance frameworks, climate change and technological and infrastructural capacity. Arctic shipping introduces various environmental challenges, including air pollution, climate forcing, marine chemical contamination, bioinvasions and animal behavioural disturbances. These impacts extend across multiple environmental spheres and can be more pronounced than in other regions owing to the extreme fragility of the Arctic environment. Overall, as climate change progresses, the economic incentives for Arctic shipping will intensify, and the pressure

for environmental sustainability will also increase. Existing governance frameworks, technologies and infrastructure have laid a foundation for sustainable Arctic shipping but remain inadequate in ensuring safety and sustainability.

Future research should better quantify how drivers of Arctic shipping interact and evolve under different socioeconomic and climate scenarios. Improved projections of Arctic shipping trends are needed to inform the development of policies to mitigate environmental damage. Numerous open-access datasets relevant to Arctic shipping are available, such as Arctic Ship Traffic Data (ASTD; <https://pame.is/ourwork/arctic-shipping/current-shipping-projects/astd/>), National Snow and Ice Data Center (NSIDC; <https://nsidc.org/data>), Coupled Model Intercomparison Project (CMIP; <https://metagrid.esgf-west.org/search/cmip6/>), Northern Sea Route Information Office (<https://arctic-luo.com/>), Canada Ice Service (<https://www.canada.ca/en/environment-climate-change/services/ice-forecasts-observations/latest-conditions.html>). These datasets are valuable for understanding Arctic shipping activities, trends in future sea ice distribution and the Arctic shipping management policies. However, given the complexity of the drivers of Arctic shipping, scientifically evaluating future Arctic shipping requires the integration of interdisciplinary approaches, including economics, management and earth sciences.

The temporal and spatial coverage of research into the impacts of Arctic shipping on oil spills, invasive species, acoustic impacts, human health and aquatic systems remains limited. Moreover, the lack of a unified data and methodological framework limits comparability among results. Furthermore, empirical observations and experimental testing are needed to more robustly quantify these impacts, establish robust dose–response relationships and identify thresholds at which shipping activities induces substantial impacts. Efforts should also be undertaken to quantify cross-sphere impacts and feedback loops, thereby clarifying the interconnections among atmospheric, oceanic, cryospheric and biological processes influenced by Arctic shipping. Such efforts include conducting targeted field campaigns to explore shipping-related impacts on Arctic environment, expanding Arctic observational networks and developing coupled Earth system models.

As concerns have been raised regarding the impacts of Arctic shipping, the IMO Polar Code and national-level regulations provide important safeguards, yet there remain limitations in the existing regulatory frameworks. Discharges of oil, garbage and sewage from ships are strictly prohibited in the Arctic under the IMO Polar Code. National and regional regulations in Canada²²¹, US Alaska²²² and Norway²²³ prohibit cruise ships from discharging greywater into Arctic waters. However, greywater is not regulated at the global level and is not addressed in the IMO Polar Code and can therefore be discharged untreated. In addition, there are numerous exemptions to the IMO's ban on the use of HFO in Arctic waters²²⁴, and the IMO Polar Code contains no mandatory provisions regarding air pollutant emissions. There are also no explicit standards or requirements for biospheric impacts such as ship strikes and acoustic impacts on marine animals. To address these gaps, efforts should focus on expanding regulations to cover greywater discharges and biospheric impacts such as ship strikes and acoustic disturbances, tightening exemptions to the use of HFO, establishing stricter air pollutant emission standards and enhancing monitoring and enforcement.

Further work is needed to increase the use and performance of technologies to mitigate the environmental impacts of shipping. Although selective catalytic reduction effectively reduces ship NO_x

emissions, most ships can comply with existing emission standards without installing such systems²⁰. Therefore, stricter fuel-quality regulations and upgraded emission standards are needed to increase the use of established technologies. In addition, the performance of these technologies should be further improved. For example, accelerating the development and adoption of alternatives to conventional antifouling paints, which rely on toxic substances such as copper, would reduce the chemical pollution associated with hull maintenance. Research is also needed to advance wastewater treatment systems and microplastic filtration technologies onboard ships, including improving removal efficiency and reliability under Arctic conditions, developing compact and energy-efficient treatment units, and integrating microplastic filtration into existing sewage systems²²⁵.

Data availability

MPI-ESM1.2-LR data used in Fig. 2a available at <https://esgf-metagrid.cloud.dkrz.de/search/cmip6-dkrz/>. Average sea ice area data used in Fig. 2b available at <https://climate.copernicus.eu/sea-ice-cover-september-2024>. IPCC data in Fig. 2b are available at https://data.ceda.ac.uk/badc/ar6_wg1/data/spm/spm_08/v20210809/panel_b. Data for Fig. 2c available at <https://usicecenter.gov/Products/Arctic-Data> and <https://arcticportal.org/shipping-portlet/shipping-routes/>. Data for Fig. 2d available at <https://oarchive.arctic-council.org/items/01ddf449-9048-4d6a-a056-65303831bb63>. Data for Fig. 2e,f available via Zenodo at <https://doi.org/10.5281/zenodo.10869014>.

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Author contributions

W.Y. and Z. Luo designed this work. W.Y. drafted the manuscript and coordinated input from all co-authors. H.L. coordinated the author team and revised the manuscript. L.P. contributed to the Environmental impacts of arctic shipping section. Z. Liu contributed to the 'Technology and infrastructure' and Economic incentives sections. Z. Luo contributed to the 'Governance' section in Economic incentives. E.R., S.H.N., F.C., N.S. and R.Y. reviewed and revised the manuscript and provided constructive suggestions in their specialized areas. X.W., Z.Y., T.H. and W.Z. contributed to the revision of the manuscript. T.H. also contributed to figure preparation. Q.Z. and K.H. provided insights into the design of this work.

Competing interests

The authors declare no competing interests.

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